Tide retrospective:
J-function hypothesis weakening (L3)

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Abstract

This retrospective records a 152-line net deletion plus an architectural unification of the J-function family. The N1 tide (generic-k J-function abstraction, merged earlier today) carried a `Continuous prior` hypothesis stronger than the mathematically-minimal one identified at the original E4 deliberation. This tide weakens the hypothesis to `AEMeasurable prior volume + ContinuousAt prior 0 + global bound` and, while the seabed is fresh, also refactors the sextic specialisation[1] to invoke the parametric form rather than carry its own independent proof. The result: the parametric file gains ∼32 lines of `quasiMeasurePreserving` plumbing; the sextic file shrinks from 280 to 96 lines as its proofs become five-line specialisations. Both Claude and GPT-5.5 Pro voted Candidate B (weaken + refactor) on architectural grounds. Single session, build green after one Mathlib-API-name correction (`Measure.quasiMeasurePreserving_smul` takes the measure as an explicit argument, not implicit). Zero `sorry`, zero `axiom`, zero `native_decide`.

1 Setting

The grammar-paper precursor sequence has three closed Lean tides: E4 (quartic J-function), L1 (sextic J-function), and N1 (generic-k parametric form). All three statements take a continuous, bounded prior $\pi : \mathbb{R} \to \mathbb{R}$. But the deliberation behind E4 had GPT-5.5 Pro identify the mathematically-minimal hypothesis class as

$$\pi \in \texttt{AEMeasurable} \;\wedge\; \pi \in \texttt{ContinuousAt}\, 0 \;\wedge\; \exists M. \forall x. |\pi(x)| \le M.$$

The stronger `Continuous prior` was taken in E4 / L1 / N1 because it was easier in Lean: composition with the scaling map $u \mapsto u/t^\alpha$ at the AE-strong-measurability site is trivial when π is continuous.

L1's retrospective explicitly flagged the weakening as a follow-up, estimating ∼50 lines of `comp_quasiMeasurePreserving` plumbing. N1's retrospective re-flagged it: *"weakening should land in N1 (`kth_*`) only — the specialisations inherit."*

This tide takes both halves:

1. Weaken the parametric form's hypothesis class.
2. Refactor the sextic specialisation to invoke the parametric form (so it inherits the weaker hypothesis).

(The quartic specialisation from E4 was never merged to `main` — it lived on the unmerged sibling branch and was superseded by N1 — so the refactor scope is just sextic.)

[1] In `Laplace/OneD/JFunctionSextic.lean` (L1, 2026-05-07).

2 The thing formalised

*For the single-monomial potential $K(w) = w^{2k}/(2k)!$ with $k \geq 1$ and a globally bounded prior $\pi : \mathbb{R} \to \mathbb{R}$ that is **AEMeasurable** and **ContinuousAt** 0,*

$$\lim_{t\to\infty} t^{1/(2k)} \int_{\mathbb{R}} e^{-t\, w^{2k}/(2k)!}\, (\pi(w) - \pi(0))\, dw \;=\; 0,$$

$$\lim_{t\to\infty} t^{1/(2k)} \int_{\mathbb{R}} e^{-t\, w^{2k}/(2k)!}\, \pi(w)\, dw \;=\; \pi(0) \cdot \tfrac{1}{k} \cdot ((2k)!)^{1/(2k)} \cdot \Gamma(1/(2k)).$$

The two parametric Lean signatures (now with the weakened hypothesis class):

```
theorem kth_jfunction_centered_tendsto_zero
    {k : ℕ} (hk : 1  k)
    {prior : ℝ → ℝ}
    (hprior_meas : AEMeasurable prior volume)
    (hprior_cont0 : ContinuousAt prior 0)
    {M : ℝ} (hprior_bd :  x, |prior x|  M) :
    Tendsto ... atTop ( 0)

theorem kth_jfunction_asymptotic
    {k : ℕ} (hk : 1  k)
    {prior : ℝ → ℝ}
    (hprior_meas : AEMeasurable prior volume)
    (hprior_cont0 : ContinuousAt prior 0)
    {M : ℝ} (hprior_bd :  x, |prior x|  M) :
    Tendsto ... atTop ( (prior 0 * ...))
```

The two sextic theorems shrink from ∼80 lines each to ∼10 lines each, becoming thin specialisations at $k = 3$ with three constant-bridging private lemmas ($(2 \cdot 3 : \mathbb{N}) = 6$ definitionally, $(6! : \mathbb{R}) = 720$ via `decide` and $1/((2 \cdot 3 : \mathbb{N}) : \mathbb{R}) = 1/6$ via `norm_num`).

3 Proof strategy

The structural argument is unchanged from N1: substitute-then-DCT with envelope $2M \cdot e^{-u^{2k}/(2k)!}$ and pointwise limit zero. Only two sites in the proof use the old `Continuous prior` hypothesis, both inside the centred theorem:

Site 1: AE strong measurability. The DCT requires strong AE measurability of the integrand $F(t, u) := e^{-u^{2k}/(2k)!} \cdot (\pi(u/t^{\alpha}) - \pi(0))$ with respect to t. The old proof used the continuous-composition upgrade. The weakened version routes through Mathlib's quasi-measure-preserving composition for the scaling map $u \mapsto (t^{\alpha})^{-1} \cdot u$, which is canned for Haar-measure spaces (and `volume` on $\mathbb{R}$ qualifies).[2] The reshape $u/t^{\alpha} = (t^{\alpha})^{-1} \cdot u$ takes the division into a scalar action, making the canned scaling lemma applicable. Then `AEMeasurable.aestronglyMeasurable` upgrades.

The eventual positivity of t matters here: `Real.rpow` on a negative base does not give the expected $t^{\alpha} \neq 0$ from $t \neq 0$, so the scaling factor's nonzero-ness needs $0 < t$. The proof was already inside an `eventually_gt_atTop` filter; the fix is local.

[2]Named `AEMeasurable.comp_quasiMeasurePreserving` with `Measure.quasiMeasurePreserving_smul`.

Site 2: pointwise limit at u. The DCT requires $F(t, u) \to 0$ as $t \to \infty$ for a.e. u. The argument $\pi(u/t^\alpha) \to \pi(0)$ comes from $u/t^\alpha \to 0$ (`Tendsto.div_atTop`) composed with `Continuous.tendsto 0` (old) or `ContinuousAt.tendsto` (new). One-line replacement.

The asymptotic theorem. Inherits the new hypothesis bundle without further proof changes (it's a corollary of the centred theorem).

The sextic specialisation. Each theorem reduces to a five-line call: instantiate `kth_*` at $k = 3$, then bridge $(2 \cdot 3 : \mathbb{N})$, $(2 \cdot 3)!$, and $1/((2 \cdot 3 : \mathbb{N}) : \mathbb{R})$ to their numeric forms via three `simp_rw`-firing lemmas.

4 Roadblocks and resolutions

One Mathlib-API-name correction during the build, two tactical adjustments, and the standard `simp_rw` pattern for the specialisation bridges.

Mathlib lemma signature. The canned scaling-quasi-MP lemma takes the measure μ as an *explicit* positional argument (it lives inside a section with an explicit `variable` $(\mu : MeasureE)), notimplicit.First-attemptcallwithoutpassing\mu$ `gave the diagnostic` *``argument has type ...* $\neq 0butisexpectedtohavetypeMe$ $--thetypemismatchwastheclue.Fix : pass\mu =$ *volume as the first argument. Five-second adjustment.*

Sextic specialisation: `convert` **versus** `simp_rw`. *First attempt at the sextic refactor used convert h using 2 to align the parametric form's Tendsto statement with the sextic-specif form. This left several minor goals that needed independent constant-bridging proofs, ending in ``*`No goals to be solved`*'' as the convert closed cleanly but the trailing tactics fired anyway. Cleaner approach: do all constant-bridging via simp_rw at h on the parametric hypothesis, then close with exact h. Three simp_rw-shaped private lemmas (for the $\mathbb{N}$-power, the factorial, and the rational inverse) suffice; the underlying integrand and limit forms align by rfl once all three are bridged.*

No GPT thrashing-rescue. *The deliberation at Step 2 named the right Mathlib lemma (quasiMeasurePreserving_smul) and the right shape; the Step 3 frictions were name-and-form bookkeeping, not strategy errors.*

5 What was Mathlib, what was new

Mathlib. *The scaling quasi-MP fact for Haar measure on finite-dim $\mathbb{R}$ vector spaces, the AE-measurability transport under quasi-MP composition, the upgrade from AE measurability to AE strong measurability for $\mathbb{R}$-valued functions, and the standard ContinuousAt.tendsto fact. All familiar; no new Mathlib API surfaced.*

Laplace seabed (existing). *N1's parametric kth_* theorems (now weakened in this tide) and the supporting private helpers (kth_jfunction_centered_subst, kth_dominator_integrable, etc.) --- unchanged. L1's sextic-specific helpers (sextic_jfunction_centered_subst, sextic_dominator_integrable) ---* `deleted` *as part of the refactor; the parametric versions subsume them.*

New. *Three constant-bridging private lemmas in the refactored sextic file (numeric specialisations of $k = 3$). The parametric file gains ~32 lines of inline quasi-MP plumbing inside kth_jfunction_centered_tendsto_zero (helper not factored out, since only one use site; if a third k-specialisation tide materialises, extracting an aemeasurable_co helper would pay off).*

6 Lessons

- The minimal-hypothesis-class deliberation pays off late, not early. *GPT-5.5 Pro identified the AEMeasurable + ContinuousAt 0 + bound class at E4's Step 2 deliberation; E4, L1, and N1 all chose the stronger Continuous prior for Lean ergonomics. This tide redeemed the deferred work; the cost was small (~32 lines of parametric-side plumbing) but the architectural payoff (-184 lines on the sextic side, sextic now inherits future parametric improvements automatically) was much larger. Lesson: the minimal hypothesis class is worth landing as a follow-up tide once the seabed is fresh, even when the original tide takes the easier-in-Lean form. Don't make a religion of it during the original tide.*

- simp_rw at h beats convert for specialisation bridging. *The sextic specialisation has the same underlying mathematical content as $k = 3$ in the parametric form; the difference is purely numeric ($(2 \cdot 3 : \mathbb{N}) = 6$ etc.). Several rounds of simp_rw at h followed by exact h is more robust than convert for this pattern, because convert leaves goals at depth-mismatch and forces extra work; simp_rw at h drives the rewrites in the right direction without leaving leftovers.*

- Mathlib lemma signatures vary in implicit/explicit bindings. *Measure.quasiMeasurePreserv takes the measure as an explicit positional argument; many other quasiMeasurePreserving_* lemmas take it implicit. This is normal Mathlib --- the explicit-binding pattern reflects the section's variable declarations --- but it's a one-attempt-then-fix cost. Worth a CLAUDE.md note for future tides.*

7 Follow-ups

- Promote aemeasurable_comp_div_const. *If a third hypothesis-weakening tide arrives (e.g., for a tide outside the J-function family that uses the same u/c scaling pattern), factor the inline plumbing of this tide into a public helper. Two use sites is currently below the refactor threshold; three would be over.*

- Weaken further to AE bound. *The current $\forall$ x. |prior x| $\leq$ M could be relaxed to $\forall^{\mu}$ x. |prior x| $\leq$ M (essentially bounded) with minor proof changes. GPT explicitly said* ``not this tide''*; mark as a refinement candidate for a future hypothesis-weakening tide.*

- \leanref-tag the grammar paper with the weaker hypothesis class. *Once the grammar paper has TeX statements, both the centred and asymptotic theorems should be pinned to this commit and tagged with the weakened-hypothesis form.*

- Promote kth_jfunction_centered_subst and kth_dominator_integrable to lean-common. *Now that the sextic file no longer has its own copies, these are the only Laplace-specifi*

J-function helpers; they're potentially useful for any monomial-Gibbs Laplace-method tide. Defer until a consumer outside laplace materialises.

Acknowledgements

*GPT-5.5 Pro's deliberation contributed three load-bearing pieces of guidance. First, the recommendation to use AEMeasurable rather than AEStronglyMeasurable as the public hypothesis (''*AEStronglyMeasurable is proof-shaped rather than theorem-shaped; recover it internally via aestronglyMeasurable*''). Second, the architectural vote for Candidate B (refactor sextic), with the practical suggestion of two-commits-one-PR packaging (collapsed here for git simplicity). Third, the Mathlib-API-name candidates and the eventual-positivity caveat for Real.rpow. Full consult preserved at the GPT v1 file.*[3]

This Tide branched off the post-O1-merge main (commit dc9d260). No in-flight tides interact with this work; the parametric file's only consumer is the sextic file, which is also modified here.

[3] Preserved at `projects/primer/tide-log/gpt55_tide_jfunction_hypothesis_weakening_v1.md`.